

Online Smart Light Detection and Monitoring System using Machine Learning

Korawit Intan
School of Information Technology
Mae Fah Luang University
6031501005@lamduan.mfu.ac.th

Nattapon Pramoonjakko
School of Information Technology
Mae Fah Luang University
6031501067@lamduan.mfu.ac.th

Theetametha Sripenbenja
School of Information Technology
Mae Fah Luang University
6031501068@lamduan.mfu.ac.th

ABSTRACT

This project presented online smart light detection by having a light sensor to send to the sensor receiver which is Zigbee. It connects with ESP32 to send data to be stored in MySQL. In addition, it can see all data and analyze it by using Machine Learning to predict the value. This method will examine the status of the light in each room that can solve the problem about forgetting to turn off the light and can analyze data of using light as well.

I. INTRODUCTION

Nowadays, the world is faced with many challenges in reducing energy consumption. Meanwhile, there are many technologies that can be used to solve these problems and support a better life by wireless sensor networks (WSN).

[1]. WSNs deployed many small sensor devices to detect environmental properties. WSNs are the combination of embedded system and wireless communication which allows data transmission among the sensor nodes over ad-hoc wireless networks as shown in Figure 1.

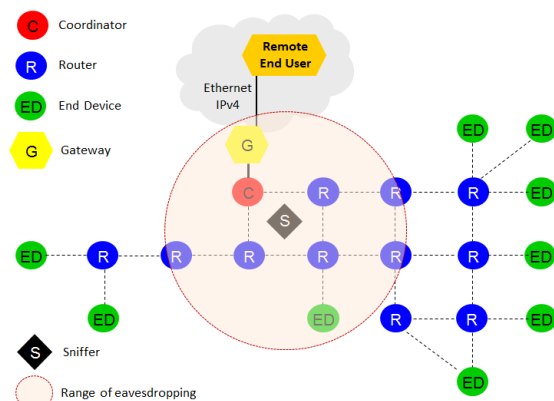


Figure 1: Wireless Sensor Networks Of Zigbee

In this prototype system, we used the TSL2561 sensor system to measure the light temperature in the lux unit. Also, we use the XBee module in terms of IEEE.802.15.4 / Zigbee Wireless Personal Area Network Standard (WPAN) to create WSN that uses low energy consumption, low maintenance, and self-organization. [2] Small size, low power, low cost and long battery life are the reasons for using ZigBee. We use the Arduino board which comes with ESP32 for easy interfacing with the ZigBee module and for easy programming (in C) of the microcontroller. The Arduino boards come with a library for interfacing with XBee modules and for dealing with analog or digital inputs and outputs. The remaining of the paper is organized as follows. In Section II, we discuss some design of the system. Section III describes the system development in this project. Then, the results will be shown in the fourth section. Finally, the conclusion of this paper is presented in Section V.

II. SYSTEM DESIGN

This system needs to use Zigbee to connect with the router that Zigbee can send data through the router. Therefore, we designed a light sensor connected with Arduino for writing code to get light values. We also connected RX, TX pins of Zigbee to send data to be stored in Zigbee by setting XCTU program. Moreover, we used 2 pieces of Zigbee to be compared as a router to be able to connect to each other. It doesn't need to use wifi because we used wireless to send data to connect the server. Importantly, we used ESP32 to connect the university server for collecting light values data throughout 24 hours. It will be stored in MySQL by writing PHP, a programming language. Lastly, we bring the data to train and test the model for predicting the status of light values.

The picture shows the system of this project.

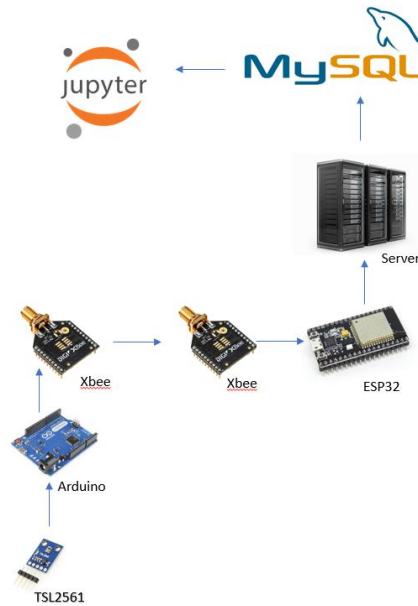


Figure 2: system design

III. SYSTEM DEVELOPMENT

The system proposed in this paper includes 3 main parts: configure Arduino, Xbee and ESP32 3 board; connect devices and programming; install server and connect everything into one web page. The following contents will describe step by step information mentioned above.

A. Configure Hardware

Firstly, the Arduino Uno board used in this paper is a microcontroller board based on the ATmega328P including 14 pins and 16 MHz quartz crystal. The board needs to be configured by the Arduino software (IDE) before it can be coded and connected with other devices. We have connected an LDR sensor which has 4 pins including 3.3V, ground, SCL and SDA. Then, it has Zigbee pins which are RX and TX pins to connect to get data from the sensor.

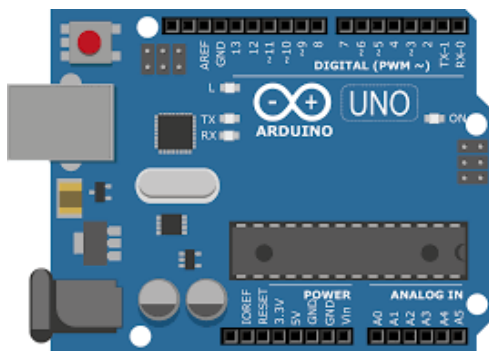


Figure 3: Arduino Uno Board

Secondly, the module [3]Xbee using in this paper is from digi xbee 3 pro embedded RF

transceiver modules to provide wireless connectivity to each other by Zigbee protocol. To configure this module, the XCTU was used and test the connection between Coordinator and Router by AT commands and MAC address. This connection can reach 60 m indoor and 1200m outdoor or RF line of sight range [2]. The 3rd picture below show the version digi xbee 3 pro of Xbee which is used in this system



Figure 4: Xbee digi 3 pro

Next, ESP32 board [4] ESP32 is a low-cost System on Chip (SoC) Microcontroller from Espressif Systems, the developers of the famous ESP8266 SoC. It is a successor to ESP8266 SoC and comes in both single-core and dual-core variations of Tensilica's 32-bit Xtensa LX6 Microprocessor with integrated Wi-Fi and Bluetooth. Furthermore, we have written ESP32 code to connect with Xbee for getting values. Then, we connected it with wifi and server to be recorded in MySQL.

Figure 5: ESP32

B. Webserver install

In this project, we used the university server to download Xampp and write a PHP program for connecting between ESP32 and MySQL. So, it can collect big data on the university server.

C. Machine Learning

We used the Jupyter program to write code(program) and have the data from MySQL to

convert CSV files for separate Attributes. Therefore, our project has 7 attributes which are Ldr, time, hour, minutes, time of days, days of week, and light in rooms but in train model we uses one hot encoding method for separation Time of days and Days of week column to numeric value another drops for Light in room which is label prediction attribute. We use KNN method for testing the model and the outcome is pleasant and the accuracy is 0.996. These are components of the prediction values. Besides, we still trained another 3 models including decision tree, Naïve Bayes, and SVM, the accuracy is 0.96, 0.961, 0.994 respectively. After we trained and tested the data, we found that accuracy of KNN value is greater than other algorithms.

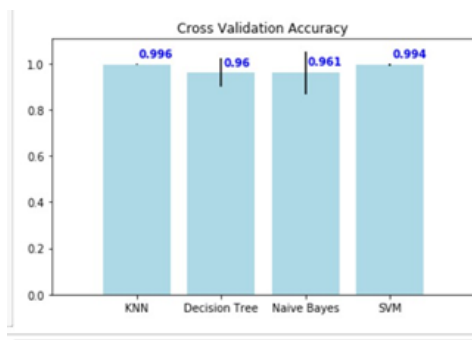


Figure 6. The performance of each algorithm

IV. RESULTS

The Online Smart Light Detection and Monitoring System using Machine Learning in this project was successful by connecting from wifi or router to connect to server. The sensor that we use is TSL2561 that can measure the light value to examine the intensity of light and record in MySQL every 1 minute throughout 24 hours. It can see these values on Website. Zigbee can communicate between coordinator and router to send data.

The figure 7 below presents the setup of this project and this operation on the web server is shown on figure 8

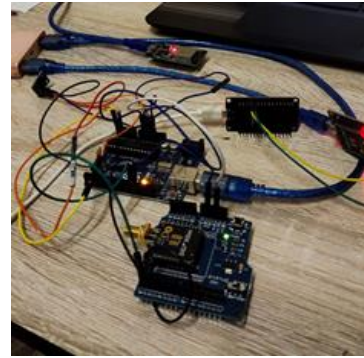


Figure 7: process of the project

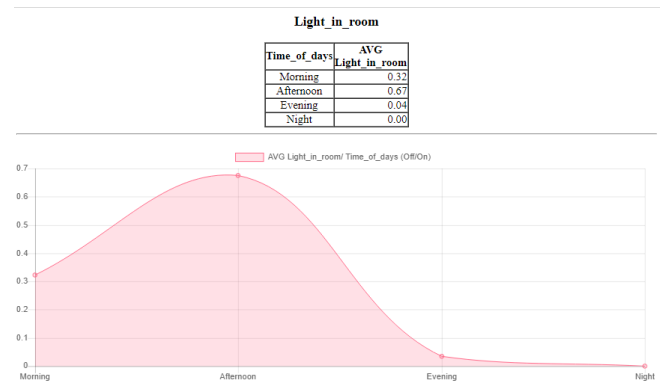


Figure 8: web server

V. CONCLUSION

This project presented the design of wireless sensor networks using ESP32, Arduino ,and Xbee for collecting data from sensors of the light measure in each place. Then, we analyzed the data by using Machine Learning that the result of comparing with KNN, Decision tree, Naïve Bayes and SVM the outcome that we use for the test is KNN model which the highest accuracy value is 0.996. Next, we connected with multiple wifi routers and showed status through the website the state of each light.

References

- 1) Poslad. (2009). Ubiquitous Computing: Smart Devices, Environments and Interactions. Retrieved from http://pervasivecomputing.se/M7012E_2014/material/Wiley.Ubiquitous.Computing.Smart.Devices.Environments.And.Interactions.May.2009.eBook.pdf
- 2) Digi International Inc. (2008). XBee ZNet2.5/XBee-PRO ZNet2.5 OEM RF Modules. Retrieved from <https://www.sparkfun.com/datasheets/Wireless/Zigbee/XBee-2.5-Manual.pdf>
- 3) Khanh. (2019). A Wireless Sensor Network for Aquaculture Using Raspberry Pi, Arduino and Xbee. Retrieved from <https://ieeexplore.ieee.org/abstract/document/8823104>
- 4) Teja. (2021, February 17). Getting Started with ESP32 | Introduction to ESP32. Retrieved from <https://www.electronicshub.org/getting-started-with-esp32/>